1. Input

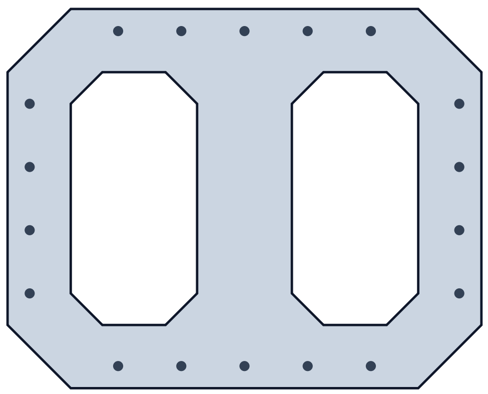
Section, materials and resistance basis exactly as analysed.

Identity

Project	**PM-advanced (7) 2D reference case**	Section	**PM-advanced (7) 2D — 1500x1200 .**
Calculation profile	**KDS 2024 — Stress–strain integratio.**	Mechanics	**Stress-strain integration**
Resistance profile	**KDS 2024 current set · resistance: K.**	Profile edition	**2024 partial-revision framework (effe.**
Verification status	**draft**	Concrete model basis	**Code-default**
Method id	**kds-2024-current-set-global-strength.**	Units	**mm · N · MPa; forces reported in kN,.**
Sign convention	**Compression positive; Mx = ΣF·(y-yc.**		

Section

Cross-section (mm, about the net concrete centroid) 18 bars



Section properties

Gross concrete area Ag	**1,120,000 mm²**
Reinforcement area As	**14,476 mm²**
Reinforcement ratio ρ	**1.293 %**
Bar count	**18**
Net centroid	**(0.0, 0.0) mm**
Bounding box	**1,500.0 × 1,200.0 mm**
Solids / holes	**1 / 2**
Perimeter	**9,262.7 mm**

Materials

Name	**KDS C30 (workbook Input)**	Standard	**KDS**
fck	**30.0 MPa**	Model	**kds-parabolic**
εcu	**0.00330**	εc0	**0.00200**
α	**0.8500**	Ec	**28,417 MPa**
Tension	**ignored**	Name	**SD400 (workbook Input)**
Standard	**KDS**	fy	**400.0 MPa**
Es	**200,000 MPa**	Model	**elastic-perfectly-plastic**
εy	**0.002000**	εu	**not declared**
Grades used	**1**		

Design resistance basis

Format	**Global resultant strength-reduction f.**	Transverse reinforcement	**other**
ϕ compression (ties)	**0.650**	ϕ compression (spiral)	**0.700**
ϕ tension	**0.850**	εt,limit	**0.00500**
Transition rule	**fixed-or-yield-multiple**	Axial cap	**0.800 × factored compression pole**

Sampling and numerical evidence

Surface points	**2400**	Directions	**96**
Stations	**25**	Direction tolerance	**0.500 %**
Worst interpolation error	**0.4077 %**	Tolerance reached	**yes**
Concrete discretization	**864 cells / 5196 points**	Strain domain	**concrete-pivot-ultimate**

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is integrated from the material stress-strain law over a verified triangle/quadrature mesh of the section.

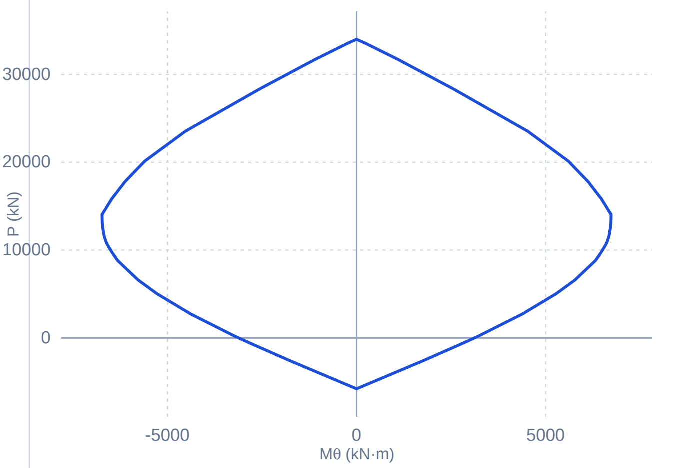
Reinforcement schedule

#	Bar id	(mm)	X (mm)	Y (mm)	As (mm²)	Steel
1	1	32.0	-400.0	530.0	804.2	SD400 (workbook Input)
2	2	32.0	-200.0	530.0	804.2	SD400 (workbook Input)
3	3	32.0	0.0	530.0	804.2	SD400 (workbook Input)
4	4	32.0	200.0	530.0	804.2	SD400 (workbook Input)
5	5	32.0	400.0	530.0	804.2	SD400 (workbook Input)
6	6	32.0	-400.0	-530.0	804.2	SD400 (workbook Input)
7	7	32.0	-200.0	-530.0	804.2	SD400 (workbook Input)
8	8	32.0	0.0	-530.0	804.2	SD400 (workbook Input)
9	9	32.0	200.0	-530.0	804.2	SD400 (workbook Input)
10	10	32.0	400.0	-530.0	804.2	SD400 (workbook Input)
11	11	32.0	-680.0	300.0	804.2	SD400 (workbook Input)
12	12	32.0	-680.0	100.0	804.2	SD400 (workbook Input)
13	13	32.0	-680.0	-100.0	804.2	SD400 (workbook Input)
14	14	32.0	-680.0	-300.0	804.2	SD400 (workbook Input)
15	15	32.0	680.0	300.0	804.2	SD400 (workbook Input)
16	16	32.0	680.0	100.0	804.2	SD400 (workbook Input)
17	17	32.0	680.0	-100.0	804.2	SD400 (workbook Input)
18	18	32.0	680.0	-300.0	804.2	SD400 (workbook Input)

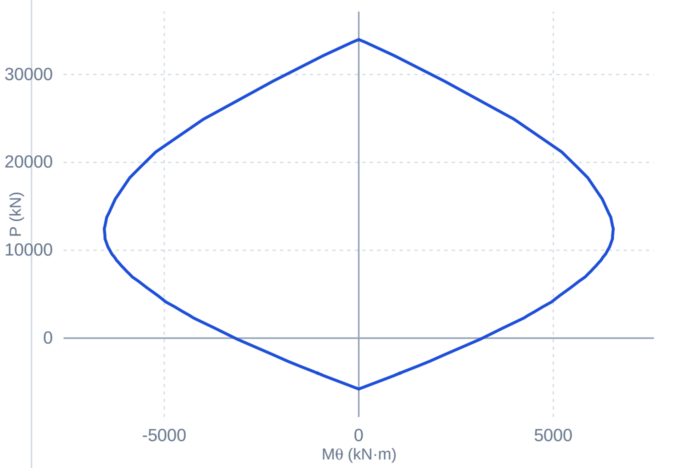
2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

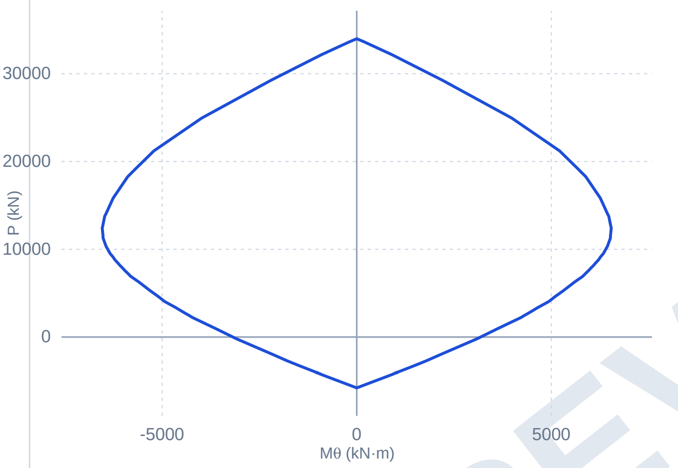
P–Mθ at θ = 0.00° Design solid · Nominal dashed



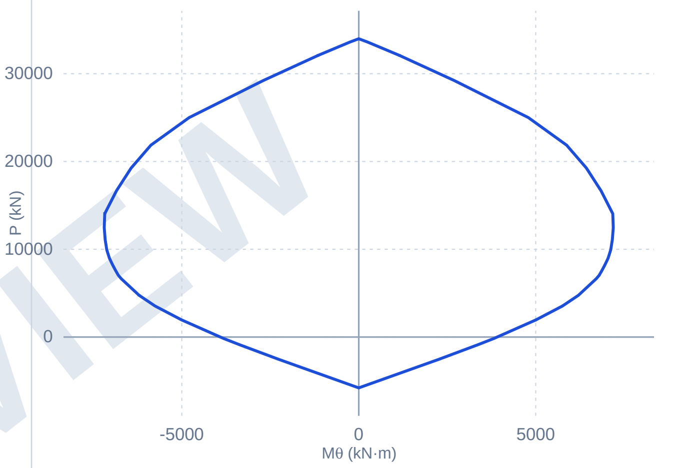
P–Mθ at θ = 20.56° Design solid · Nominal dashed



P–Mθ at θ = 21.08° Design solid · Nominal dashed



P–Mθ at θ = 90.00° Design solid · Nominal dashed



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	Pu (kN)	Mux (kN·m)	Muy (kN·m)	θL (°)	ϕ	Class	UR	Verdict
1	ULS-1 workbook Newton d.	24942.9	3714.2	1431.8	21.1	—	—	1.000	ADEQUATE
2	ULS-2 same axial, half mo.	24942.9	1857.1	715.9	21.1	—	—	0.861	ADEQUATE
3	ULS-3 low axial, high mom.	5000.0	4000.0	1500.0	20.6	—	—	0.744	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

4.1 ULS-1 workbook Newton demand (on the 15 deg surface)

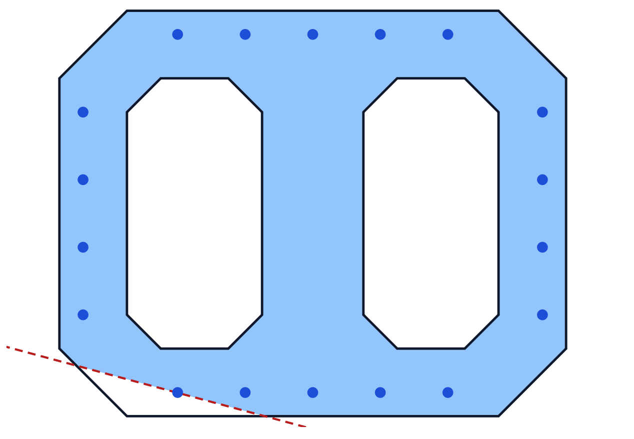
Pu = 24942.9 kN · Mux = 3714.2 kN·m · Muy = 1431.8 kN·m · θL = 21.08°

ADEQUATE · utilization 0.9998 · fixed-P diagnostic 0.9993

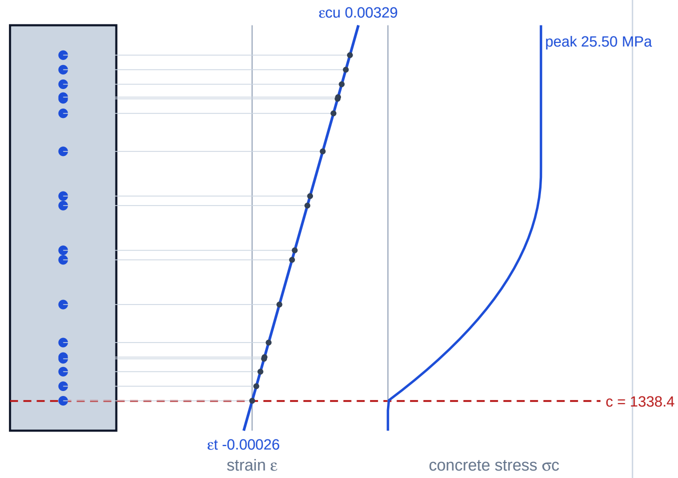
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone ε>0



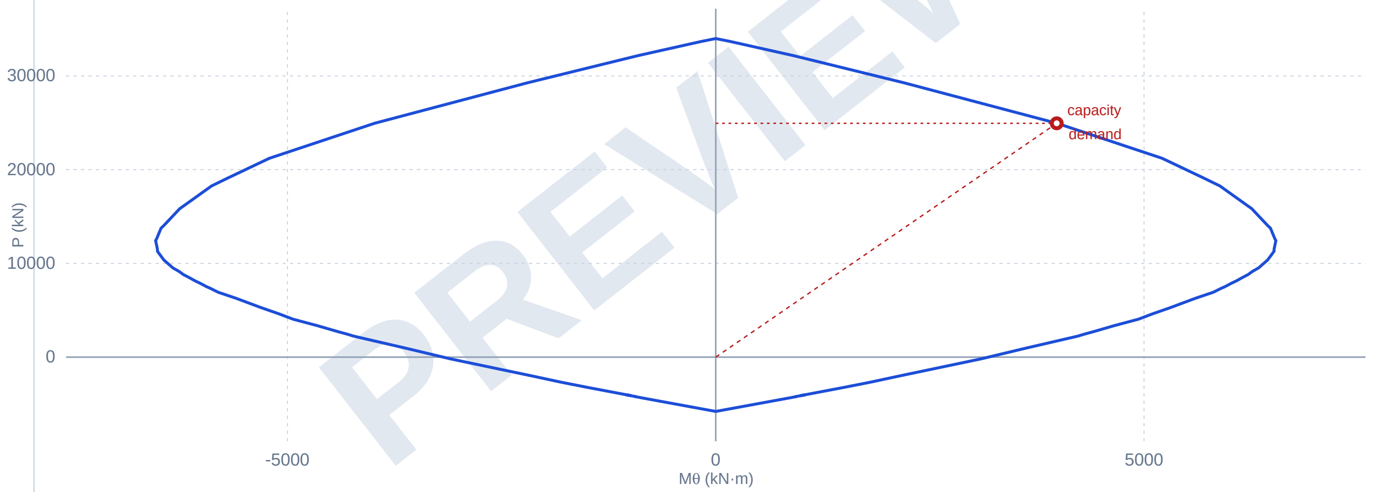
Longitudinal section — strain and stress over the depth Dθ=1444



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction

P–Mθ at θL = 21.08°, with the load point solid = Design · dashed = Nominal · the dotted ray is the governing proportional check



Concrete and resultants

Concrete force	**21,577.4 kN**	Concrete Mx	**2,900.4 kN·m**
Concrete My	**1,122.3 kN·m**	Steel force (gross)	**3,629.5 kN**
Displaced concrete	**-264.0 kN**	Steel net	**3,365.5 kN**
Response P	**24,942.9 kN**	Response Mx	**3,714.2 kN·m**
Response My	**1,431.8 kN·m**	Residual |ΔP|	**1.01e-4 N**
Residual |ΔMx|	**3.15e-3 N·mm**	Residual |ΔMy|	**4.99e-3 N·mm**

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.8	2.520e-3	400.00	321.70	—
2	-200.0	530.0	804.2	262.0	2.648e-3	400.00	321.70	—
3	0.0	530.0	804.2	210.1	2.775e-3	400.00	321.70	—
4	200.0	530.0	804.2	158.3	2.903e-3	400.00	321.70	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.5	3.030e-3	400.00	321.70	—
6	-400.0	-530.0	804.2	1337.6	1.971e-6	0.39	0.32	—
7	-200.0	-530.0	804.2	1285.8	1.295e-4	25.89	20.82	—
8	0.0	-530.0	804.2	1233.9	2.570e-4	51.39	41.33	—
9	200.0	-530.0	804.2	1182.1	3.845e-4	76.89	61.84	—
10	400.0	-530.0	804.2	1130.3	5.120e-4	102.39	82.35	—
11	-680.0	300.0	804.2	608.5	1.795e-3	359.09	288.80	—
12	-680.0	100.0	804.2	801.7	1.320e-3	264.06	212.37	—
13	-680.0	-100.0	804.2	994.8	8.451e-4	169.02	135.94	—
14	-680.0	-300.0	804.2	1188.0	3.699e-4	73.99	59.50	—
15	680.0	300.0	804.2	256.1	2.662e-3	400.00	321.70	—
16	680.0	100.0	804.2	449.2	2.187e-3	400.00	321.70	—
17	680.0	-100.0	804.2	642.4	1.712e-3	342.42	275.39	—
18	680.0	-300.0	804.2	835.6	1.237e-3	247.38	198.96	—

Solver and admissibility evidence

Strain plane ε0	**1.5162e-3**	Curvature kx	**2.3759e-6**
Curvature ky	**6.3748e-7**	Neutral-axis line	**164.9806°**
Compression depth c	**1,338.387 mm**	Iterations	**7**
Relative residual	**2.576e-12**	Max concrete compression	**0.003292**
Concrete limit	**0.003300**	Max steel tension	**0.000000**
Steel limit	**not declared**	Strain admissibility	**admissible**
Governing utilization	**0.9998**		

4.2 ULS-2 same axial, half moment (inside)

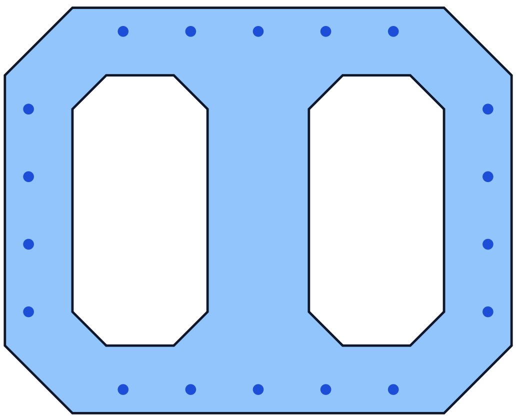
Pu = 24942.9 kN · Mux = 1857.1 kN·m · Muy = 715.9 kN·m · θL = 21.08°

ADEQUATE · utilization 0.8606 · fixed-P diagnostic 0.4996

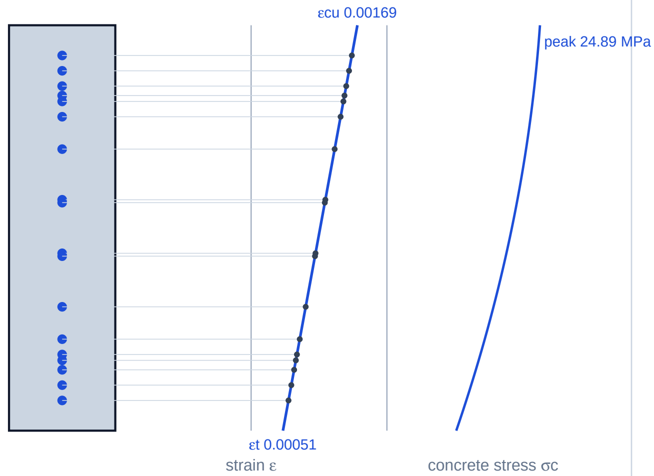
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone ε>0



Longitudinal section — strain and stress over the depth Dθ=1456

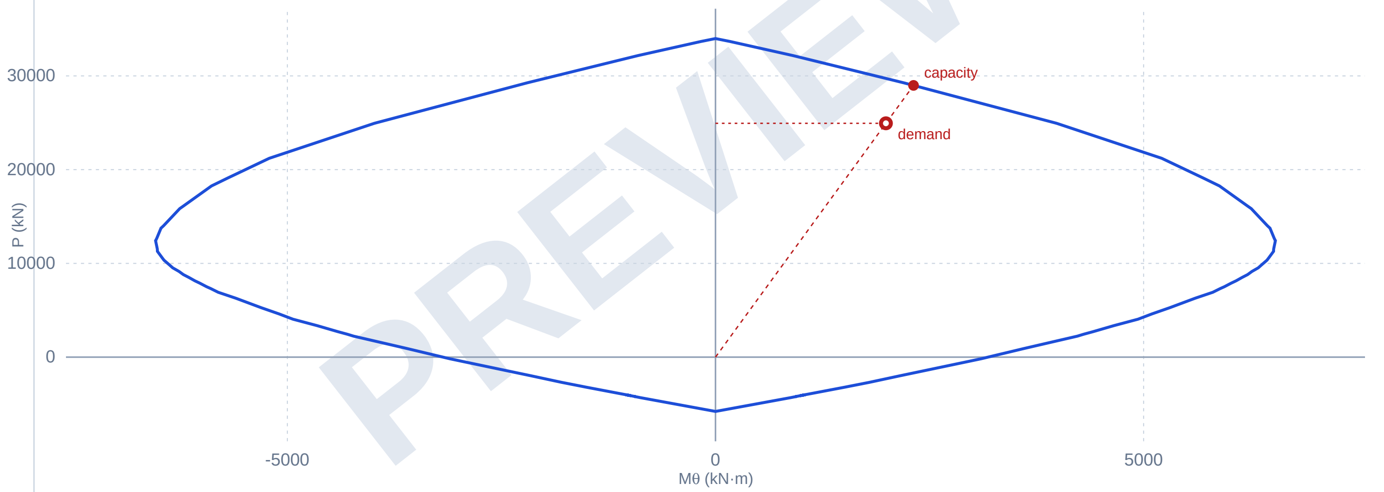


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction

c = 2079.0

P–Mθ at θL = 21.08°, with the load point solid = Design · dashed = Nominal · the dotted ray is the governing proportional check



Concrete and resultants

Concrete force	**22,045.7 kN**	Concrete Mx	**1,476.8 kN·m**
Concrete My	**563.3 kN·m**	Steel force (gross)	**3,180.1 kN**
Displaced concrete	**-282.9 kN**	Steel net	**2,897.2 kN**
Response P	**24,942.9 kN**	Response Mx	**1,857.1 kN·m**
Response My	**715.9 kN·m**	Residual |ΔP|	**1.43e-5 N**
Residual |ΔMx|	**5.19e-3 N·mm**	Residual |ΔMy|	**1.92e-3 N·mm**

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	328.6	1.423e-3	284.64	228.92	—
2	-200.0	530.0	804.2	273.6	1.468e-3	293.59	236.12	—
3	0.0	530.0	804.2	218.6	1.513e-3	302.54	243.32	—
4	200.0	530.0	804.2	163.6	1.557e-3	311.49	250.51	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	400.0	530.0	804.2	108.6	1.602e-3	320.43	257.71	—
6	-400.0	-530.0	804.2	1347.7	5.946e-4	118.92	95.64	—
7	-200.0	-530.0	804.2	1292.7	6.393e-4	127.87	102.84	—
8	0.0	-530.0	804.2	1237.7	6.841e-4	136.81	110.03	—
9	200.0	-530.0	804.2	1182.7	7.288e-4	145.76	117.23	—
10	400.0	-530.0	804.2	1127.7	7.735e-4	154.71	124.42	—
11	-680.0	300.0	804.2	626.8	1.181e-3	236.16	189.93	—
12	-680.0	100.0	804.2	819.1	1.024e-3	204.89	164.78	—
13	-680.0	-100.0	804.2	1011.4	8.681e-4	173.62	139.63	—
14	-680.0	-300.0	804.2	1203.7	7.118e-4	142.35	114.49	—
15	680.0	300.0	804.2	252.7	1.485e-3	297.00	238.86	—
16	680.0	100.0	804.2	444.9	1.329e-3	265.73	213.71	—
17	680.0	-100.0	804.2	637.2	1.172e-3	234.46	188.57	—
18	680.0	-300.0	804.2	829.5	1.016e-3	203.19	163.42	—

Solver and admissibility evidence

Strain plane ε0	**1.0984e-3**	Curvature kx	**7.8173e-7**
Curvature ky	**2.2369e-7**	Neutral-axis line	**164.0318°**
Compression depth c	**2,079.010 mm**	Iterations	**5**
Relative residual	**3.635e-13**	Max concrete compression	**0.001690**
Concrete limit	**0.003300**	Max steel tension	**0.000000**
Steel limit	**not declared**	Strain admissibility	**admissible**
Governing utilization	**0.8606**		

4.3 ULS-3 low axial, high moment

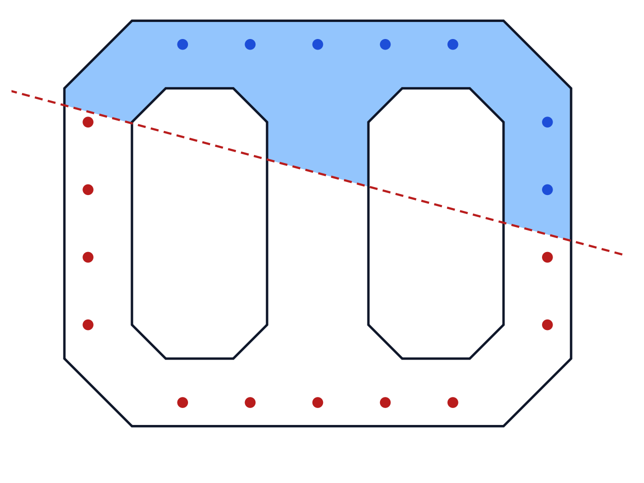
Pu = 5000.0 kN · Mux = 4000.0 kN·m · Muy = 1500.0 kN·m · θL = 20.56°

ADEQUATE · utilization 0.7442 · fixed-P diagnostic 0.8193

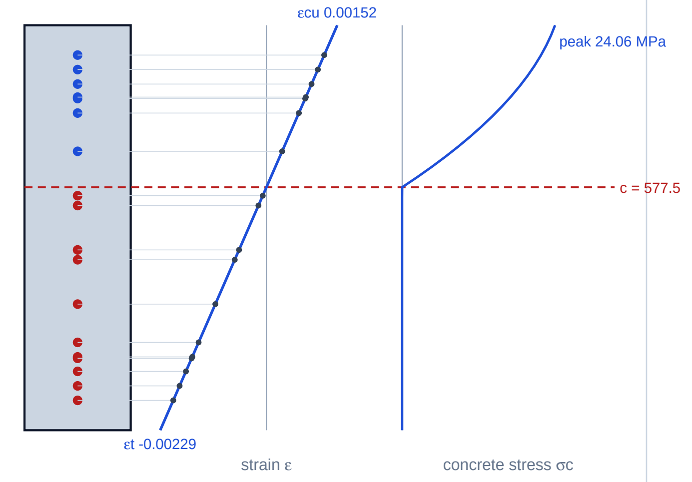
Converged preview equilibrium inside the material domain.

Section state

Transverse section — neutral axis and compression zone ε>0



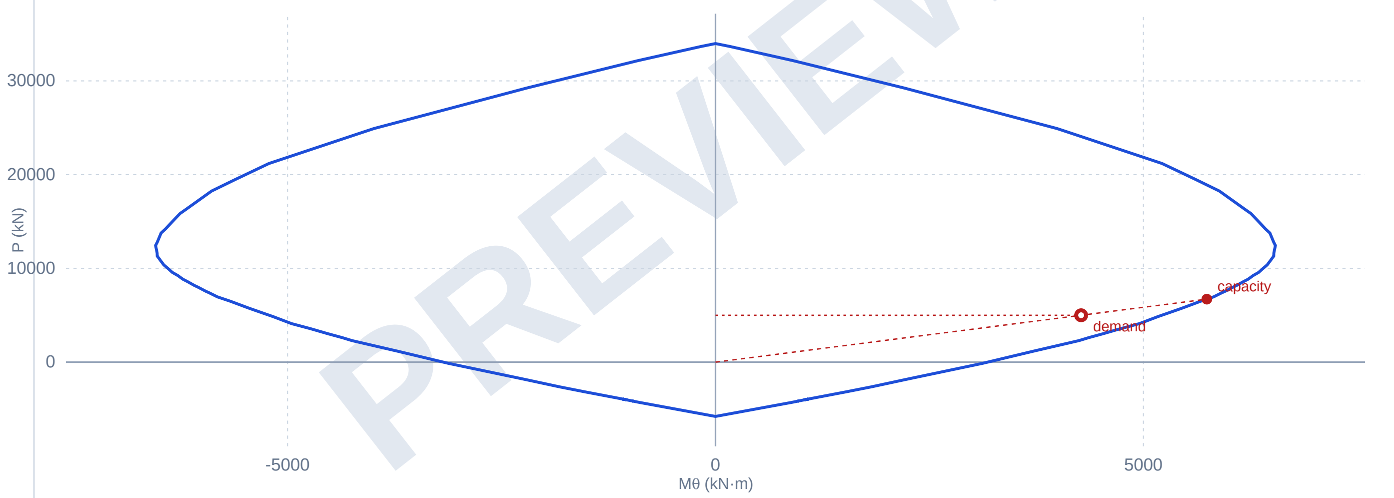
Longitudinal section — strain and stress over the depth Dθ=1443



Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Interaction diagram through the demand direction

P–Mθ at θL = 20.56°, with the load point solid = Design · dashed = Nominal · the dotted ray is the governing proportional check



Concrete and resultants

Concrete force	**6,195.6 kN**	Concrete Mx	**2,728.2 kN·m**
Concrete My	**1,023.3 kN·m**	Steel force (gross)	**-1,101.1 kN**
Displaced concrete	**-94.5 kN**	Steel net	**-1,195.6 kN**
Response P	**5,000.0 kN**	Response Mx	**4,000.0 kN·m**
Response My	**1,500.0 kN·m**	Residual |ΔP|	**5.59e-9 N**
Residual |ΔMx|	**0.00e+0 N·mm**	Residual |ΔMy|	**9.54e-7 N·mm**

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
1	-400.0	530.0	804.2	313.0	6.983e-4	139.66	112.33	—
2	-200.0	530.0	804.2	261.4	8.347e-4	166.94	134.26	—
3	0.0	530.0	804.2	209.7	9.711e-4	194.22	156.20	—
4	200.0	530.0	804.2	158.0	1.108e-3	221.50	178.14	—

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	400.0	530.0	804.2	106.4	1.244e-3	248.78	200.08	—
6	-400.0	-530.0	804.2	1337.1	-2.005e-3	-400.00	-321.70	—
7	-200.0	-530.0	804.2	1285.4	-1.869e-3	-373.74	-300.58	—
8	0.0	-530.0	804.2	1233.7	-1.732e-3	-346.46	-278.64	—
9	200.0	-530.0	804.2	1182.1	-1.596e-3	-319.18	-256.70	—
10	400.0	-530.0	804.2	1130.4	-1.460e-3	-291.91	-234.76	—
11	-680.0	300.0	804.2	607.6	-7.922e-5	-15.84	-12.74	—
12	-680.0	100.0	804.2	800.8	-5.893e-4	-117.86	-94.79	—
13	-680.0	-100.0	804.2	994.0	-1.099e-3	-219.88	-176.84	—
14	-680.0	-300.0	804.2	1187.2	-1.609e-3	-321.89	-258.88	—
15	680.0	300.0	804.2	256.2	8.483e-4	169.65	136.44	—
16	680.0	100.0	804.2	449.4	3.382e-4	67.64	54.40	—
17	680.0	-100.0	804.2	642.7	-1.719e-4	-34.38	-27.65	—
18	680.0	-300.0	804.2	835.9	-6.820e-4	-136.40	-109.70	—

Solver and admissibility evidence

Strain plane ε0	**-3.8060e-4**	Curvature kx	**2.5504e-6**
Curvature ky	**6.8197e-7**	Neutral-axis line	**165.0295°**
Compression depth c	**577.545 mm**	Iterations	**6**
Relative residual	**1.419e-16**	Max concrete compression	**0.001525**
Concrete limit	**0.003300**	Max steel tension	**0.002005**
Steel limit	**not declared**	Strain admissibility	**admissible**
Governing utilization	**0.7442**		